

Hybrid Cryptosystem based Security in Cloud Computing

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Abstract—The advancement in cloud computing has led to emergence of the era which projects the evolution in technology which is infrastructure less architecture, without actually installing the software on the end and many other development models. This is improvement from the earlier mainframe computer to a client server model which includes the benefits from grid computing, autonomic computing, and utility computing. The advantages of cloud come with the challenges and one of the most important is security. The security of resources is at risk and also the traditional security approaches is not effective for the cloud. As a result the paper discusses the potential security requirements and further will discuss the possible security solutions. The paper proposes a mechanism which is based on third party which is trusted for assuring security in cloud environment. The proposed technique is based on cryptography and public key infrastructure with single sign in and LDAP. This mechanism helps to ensure authentication, confidentiality, and integrity of concerned data and communication

Keywords—Cryptosystem, LDAP, Cloud Computing, SaaS;

I. INTRODUCTION

In 1960, time sharing utility has been introduced which is followed by networking of computers in 90s, further grid system of last decade. This kind of separation is trending now a days, as most of the academicians and entrepreneurs are attracted towards this new technology.

The Cloud computing is considered as future technology based on innovative information system. Cloud has declined the dependence of users on the hardware which reduces the complexity and client side requirement. Though the cloud is showing promising results but it has challenges on the security front. As the complexity and feature of cloud differs from traditional approaches therefore the new features and upgradation is necessary. In the paper, we tried to identify the security issues faced by clarity. The requirement for the security is discussed. The paper proposes a solution which is

based on trusted third party mechanism. Third party considered a federation of clouds for realizing the actual benefit from the information system [1,2].

In early 90's grid computing has been successfully emerged. Grid computing is a distributed architecture of large numbers of computers connected to solve a complex problem and data intensive scientific calculations. It is server that runs independent tasks and is loosely linked by the Internet or low-speed networks. On the other hand, utility computing presents the model of providing resource-on-demand and charging customers based on usage. It is a computing business model in which the provider owns, operates and manages the computing infrastructure and resources, and the subscribers access it as when required on a rental or metered basis. Cloud computing has been resulted as the combination of above techniques, which is defined as technique in which the resources are deployed on common external storage [3,4]. Cloud computing allows users with various computing capabilities to store and process data in a private cloud located in a data center, thus making data-accessing mechanisms more efficient and reliable. It often refers to the delivery of on-demand computing resources, everything from applications to data centers, over the internet on a pay-for-use basis. The basic backbone which empowers the cloud technology is virtualization technique from 1970's, which is for these years only available at mainframe systems. In the prototype, there is an application which runs on the host computer called as hypervisor which will generate virtual machines, for the simulation of physical computers ranging from operating system to end user computers. On the hardware level, different resources like hard drives, processors, are located in the data centers which are necessary for storing the data and processing the information. For all this, various software, virtualization layer and task management layer play an important role. Virtualization is the most important part of cloud execution and is used for providing cloud features like resource pooling,

access and location independence. Cloud computing uses the features of virtualization by separating logical from the physical for handling large traffic, makes the cloud better than Grid computing. In contrast to traditional networks, cloud offer flexibility and robustness. In the last management layer protects the system from different rates of traffic with the help of adding and removing servers. This layer also has the capability to secure the cloud.[5]

In Cloud various services models are defined as below:

A. INFRASTRUCTURE AS A SERVICE(IAAS):

IAAS refers to on-demanded provisioning of infrastructure resources like Processor, memory, networks, and other resources. It allows users to use different software's like operating systems, and any other application software's. Usually in terms of UMS the cloud owners who officers IAAS is called IAAS provider exp-Amaon EC2.

B. PLATFORM AS A SERVICE(PAAS)

PaaS delivers the user ability to post onto the cloud any of the consumer developed or acquired application using programming Language and tools. In this service user does not manage any of the cloud infrastructure like servers, storage, Operating System, but can manage the deployed application.

C. SOFTWARE AS A SERVICE(SAAS)

SAAS refers to providing on-demand application over the internet. User can access any application which is running on a cloud infrastructure. These applications can be accessed through client server interface like web browser. The user cannot manage the cloud infrasture in this model. For example:Rockspace

Deployment models used by different Clouds are given below:

- **Private cloud:** In Private Cloud, infrastructure is managed and controlled by private organization. It is managed by organization itself or third party organization and may be located on or off the premises of company.
- **Public cloud:** In Public Cloud, infrastructure has been made available for the enterprises and general public. The cloud in this is basically owned by the company which is selling cloud[7].
- **Community cloud:** In community Cloud, infrastructure is made available of various community based on different criteria's and applications eg., Security, policy, health, military.
- **Hybrid cloud:** in Hybrid Cloud, infrastructure is a combination of one or two clouds which in itself exist as separate entities but are tied by systemize technology, which allow data and resources Portability.

Cloud computing service will provide lot of benefits:-

- **Easy implementation:** As distributed computing administrations enable customers to explore the cloud to scene, execute the most proper cloud techniques for

speediest outcomes and give full-stack cloud applications and framework administration administrations without breaking a sweat of utilization and client benefit.

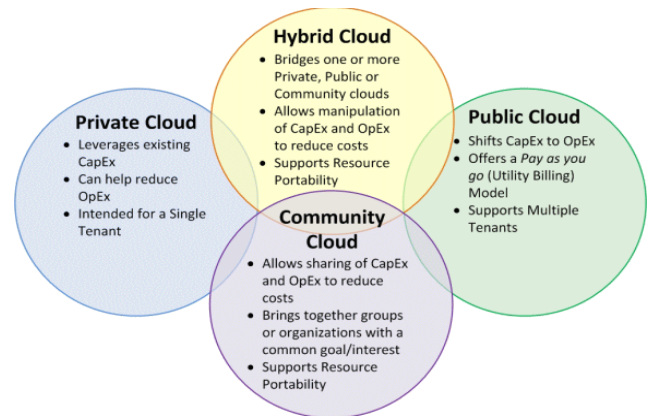


Fig 1. Types of Cloud

- **Easy accessibility:** The files are stored at location which is remote area on hard drive or cloud. The data can be accessed using the web services.
- **Scalability:** A standout amongst the most critical attributes of distributed computing is that it's adaptable. The capacity to extend and contract foundation assets on request is the thing that makes the cloud so capable.
- **Infrastructure needs :** The dat to be accessed need to be stored at remote location. To access the data it needs lot of resources. It can access the remote server using various networks like LAN, WAN, or WWW.

Most of the cloud executions have highly advanced security techniques [8,9], due to centralization of data and resources at one location. As most of the data are homogeneous therefore cloud administrators can focus on making cloud architecture secure. Further automation abilities in the cloud, improves the cloud security. Still there is a lot of identified threats which need consideration.

II. CLOUD COMPUTING SECURITY

For Securing the Information cloud system, there involves a lot of unique threats and issues which need consideration. There are various techniques, one of the promising technique is trust Based security measure. Trust has a different interpretation of a different course and specialty. Any entity is considered to be trustworthy if it is performing what it is expected and required to perform. This is called by different definitions namely, reliability and data authenticity. The trust in any institution is defined to be user or customer faith in the organization or institution that it will provide the required services very efficiently and securely with all the moral integrity. Security is defined to be the situation in which the risk of attack is either minimized or fully eliminated.

Trust in cloud computing determined mostly on the basis of which environment it is deployed that is how data and applications are managed. Earlier in traditional techniques, trust can be achieved using various security policies which add various kinds of constraint on programs. In cloud environments, this type of control is not possible. Every cloud deployment has different kind of constraints based on consumer and the owner of the cloud. For example, in case of Public or community cloud, the management will be in the hands of organization which is providing the service. In various clouds, the owner provides security policy to confirm that risk is reduced. But as this involve the biggest risk of trusting the security processes implemented by cloud owner.

The cloud environment cannot be protected using perimeter security. Perimeter security is defined to be the set of physical and programmatic security rules that protects the system from different outside malicious activities. Conventionally it is presumed that any connectivity of system or organization to outside will make the system vulnerable to threats[10,11]. These security parameters protect the system from the outside world. But in case of cloud everything is outsourced or has been placed on cloud which needs further consideration on the security front. Therefore this paper proposes security technique using trusted third party mechanism by using trust and cryptography for achieving integrity, confidentiality, and authenticity of data. The relying party will trust the TTP for the security support with acknowledgement of some minimum associated risk. By adding the TTP can particularly address the loss of traditional security domain.

III. SECURITY THREATS

For making the information system secure, various kinds of threats and challenges which needs to identified and addressed by executing various countermeasures. Lastly, the required security and various controls will be introduced to make system functional as well as efficient with the features of reliability, maintainability, etc. Cloud computing because of its architectural property and features will provide various security benefits like centralization of security, which will protect from various traditional risks but it has other challenges. In general, the basic security measures like confidentiality, availability, and integrity should be achieved for three types of resources that is data, software, and hardware. These challenges have been explained properly in section below:

A. Cloud Security

We can assure that by employing third party which is trustworthy will help us to form the system which delivers confidentiality, integrity and authenticity of data and other communications. The TTP is a system which provides different services which are delivered across different domains. The trusted third party used to acknowledge every transaction between the parties. This kind of intermediate party generates a digital certificate which is distributed to

participating servers to provide authority to the servers. TTP are connected to the chains of trust called as certificate paths which is known as public key infrastructure. it will help to provide authentication, data integrity, data confidentiality, Non repudiation, and authorization. PKI[12] is coupled with directory which is a set of objects organized in a hierarchical manner.

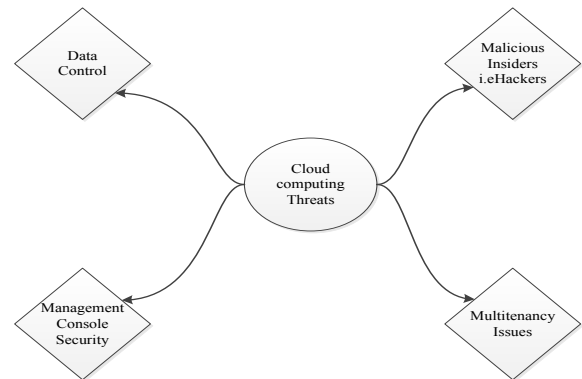


Fig 2. Cloud Computing Threats

These directories help to allocate certificates, certificate keys and private keys have been sent. The infrastructure develop one sign on mechanism where u need only once to sign the documents as the user navigate a lot out of cross boundary organizations.

- **Authentication:**

In cloud the certified authority is required to certify the all the entities involved like servers, users, network devices, etc. the public key infrastructure will provide strong credentials to all the physical and virtual entities involved in cloud. Digital signature in combination with single sign on and light directory access protocol provides the strongest authentication process for the cloud. Since the cloud is becoming the basic requirement for every service. The organizations are becoming more and more open as the data and resources are outsourced to different platforms therefore the single sign on becomes an important issue. There are different techniques for this which uses TTP LDAP[13] directory for authentication and authorization. To improve the communication between various parties, standard SAML is used that is security assertion Markup Language, which is XML based for authentication and authorization of data and resources. Shibboleth and SAML go hand in hand for privacy, business process.

- **Certificate based authentication:**

The cloud in itself is a kind of open platform where several domains can access storage and resources. The relationship between resources and user are so

dynamic and adhoc, therefore there should be a proper certification for the users. Public key infrastructure will provide the certification that will act as trust identification criteria.

- *Generation of security domains:*
Federation with public key infrastructure and LDAP technology will try to generate trust relationships between various entities. In this federated clouds are formed which are collection of different clouds which can be interoperable and data can be exchanged between them. These clouds can interoperate using one of the interfaces. This type of interface provides a structure and legal authority which help in authentication and authorization.

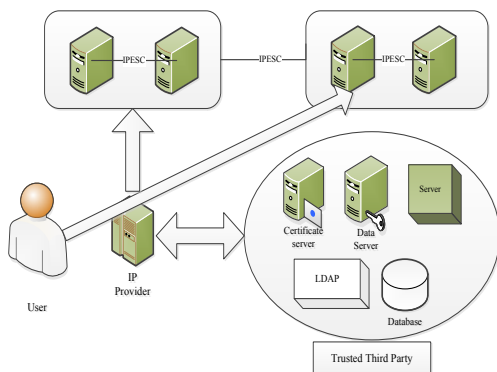


Fig 3. Authentication Process for Trusted party system

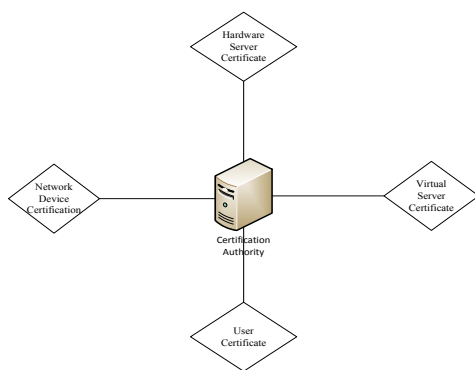


Fig 4. Different type of Certificates

- *Data Separation using different encryption techniques:*
The information which is personal or crucial in cloud is motivating factor for the deployment of SaS and AaS models. The data are encrypted using symmetric or asymmetric key cryptography in cloud in such a way that they appear as invisible to the outsider world.
- *Confidentiality:*

Confidentiality refers to only legitimate persons or organizations are able to obtain the protected data. With increase in data access by organizations or users will led to more risk. The data over the network is highly vulnerable which amounts to data modification and data interruption in the process. Public key interruption implements IPsecurity or SSL for data communication. IP layer sends the IP protected packets using TCP, UDP or ICMP, etc. IPsec adds confidentiality or authenticity. SSL add end to end encryption within various client server applications.

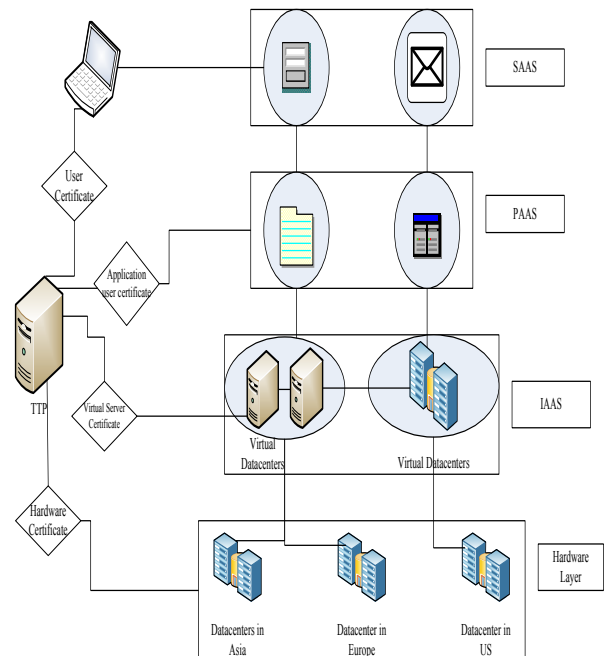


Fig 5. Trust to be developed in Topdown Fashion

The paper proposes a solution to lot of challenges in a cloud environment which makes users secure with the help of trusted third party. Trust actually works in fashion where security operates in top down fashion, as every layer needs to trust the layers below it for the reliable and secure communication. Trust certificates have been issued for assigning identity and credentials for the users. The process involved the issuance of certificates. Trusted third party is the security aspect in cloud environment with no earlier knowledge for establishing secure interactions. The users are required to authenticate using the digital certificate with in a cloud service. The certificate is issued with other service provider certificate (PaS or IaS level) to form a secure SSL connection. The client can scramble every single individual datum put away on the cloud to counter beforehand recognized classification dangers. As cloud framework's host various administrations, a few applications can be mounted on a virtual server, each requiring separate advanced authentications for SSL correspondences (distinctive ports can be utilized to help more than one SSL associations

with a virtual server). The application supplier can utilize his own testament to verify himself in correspondences with the cloud yet in addition utilize this endorsement to encode and decode application information. These testaments can be improved to convey part data about a client or process (expanded X.509 authentications).

At the most reduced level the equipment foundation proprietor makes utilization of a computerized testament to impart security amongst gadgets and virtual servers yet in addition for verification purposes if required. Key administration is a basic issue in cloud frameworks, as the virtualisation of administrations darkens the ID of the physical key stockpiling area, handicapping customary security systems. The proposed arrangement calls upon cryptography, particularly Public Key Infrastructure, to guarantee the confirmation, honesty and privacy of included information and correspondences. A TTP[14] is entrusted with guaranteeing particular security qualities inside a cloud domain, while understanding a confide in work between included elements, framing leagues of trusts. The arrangement, displays an even level of administration, accessible to every single ensnared element, that understands a security work, inside which fundamental trust is kept up. This approach influences utilization of a blend of Public Key Cryptography,

their consolidated usage. The trusted outsider can be depended upon for: Low and High level privacy, Server and Client Authentication, Generating Security Domains.

IV. CONCLUSION

Definitely cloud computing support a large amount of data though it has lot of challenges. The cloud management offers different types of cloud arrangement based on need. This new technology processing provides arrangement design, with the ability to address different susceptibility perceived in contemporary Information Systems yet its dynamic attributes can impede the capacity of customary countermeasures. In this paper we have recognized nonexclusive outline standards of a cloud situation which originate from the need to control important vulnerabilities and dangers. To do as such, software and data frameworks configuration approaches were received. Security in a cloud domain requires a fundamental perspective, from which security will be developed on confide in, moderating insurance to a trusted outsider. Public key Infrastructure, LDAP and SSO can resolve various issues in distributed computing managing the uprightness, classification, realness and accessibility of information and correspondences. The arrangement, displays an even level of administration, accessible to every single involved substance, that understands a security work through leagues, inside which basic trust is kept up.

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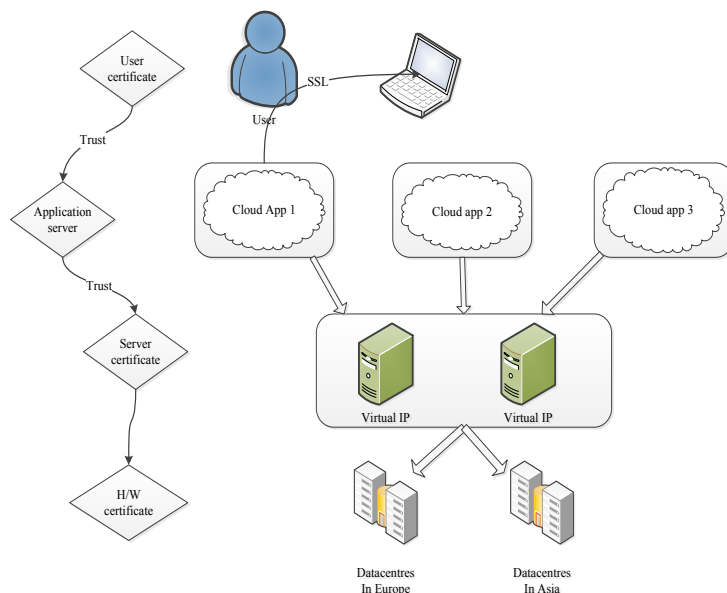


Fig 6. A client confirms himself with a cloud benefit utilizing his own testament which in mix with the specialist organization declaration is utilized to secure and scramble all correspondences.

To single Sign-On innovation and LDAP registries to safely distinguish and confirm embroiled elements. The model exhibited in this paper offers the benefits of each single innovation utilized and manages their inadequacies through

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